

Constructible Failures of the Erdős-Volkmann-Problem for Rings

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Question (Erdős-Volkmann-Ring Problem)

Is there a subring R of $(\mathbb{R}, +, \times, 0, 1)$ such that $\dim_H(R) \neq 0, 1$?

Question

What does this mean?

Descriptive set theory can answer questions about [provability](#).

Fact

Every subset of $\mathbb{R}$ has a Hausdorff dimension.

This dimension can be computed via **prefix-free Kolmogorov K complexity** and **information density**: for $x \in 2^\omega$ define

$$\dim(x) = \liminf_{n \rightarrow \infty} \frac{K(x \upharpoonright_n)}{n}.$$

J. Lutz and N. Lutz proved the following general identification:

Theorem (J. Lutz, N. Lutz (2018))

For every $A \subseteq \mathbb{R}$ we have

$$\dim_H(A) = \min_{B \in 2^\omega} \sup_{x \in A} \dim^B(x).$$

Definition

An uncountable set $A \subseteq \mathbb{R}$ has the **perfect set property** (PSP) if it contains a non-empty perfect subset.

Question

Is there an uncountable set $A \subseteq \mathbb{R}$ which does not contain an uncountable subset?

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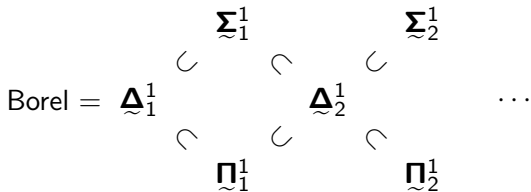
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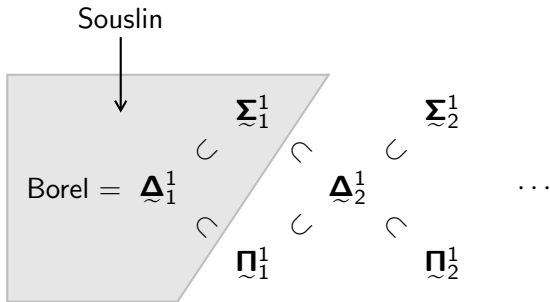
Answer

It's complicated.

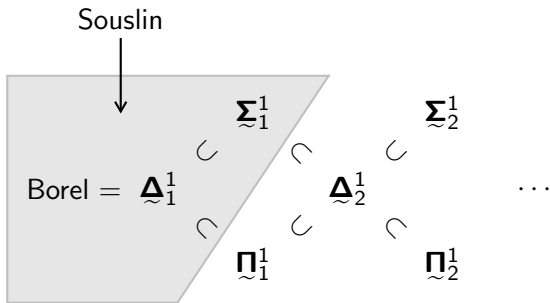
Axioms	Behaviour
ZF + DC	
ZFC	
ZF + DC + AD	
ZFC + (V=L)	



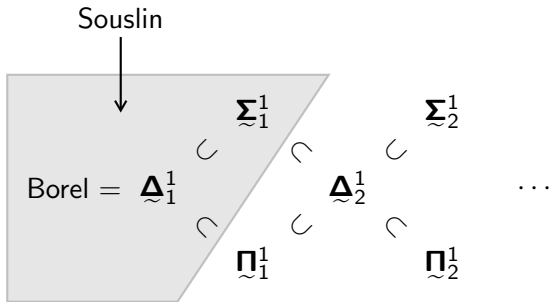
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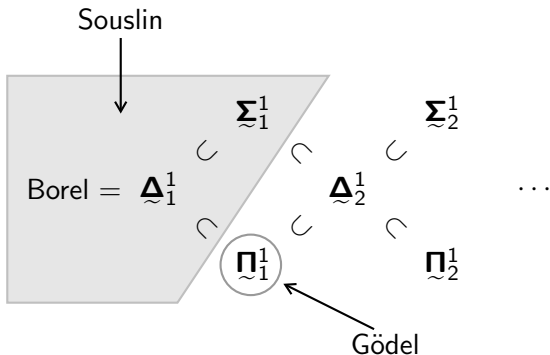
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ZFC + (V=L)	PSP fails for some Π_1^1 set (Gödel)



Theorem (P. Erdős, B. Volkmann, 1966)

For every $\alpha \in (0, 1)$ there exists a Borel *subgroup* $G \subseteq (\mathbb{R}, +)$ for which

$$\dim_H(G) = \alpha.$$

This means, G as a subset of $\mathbb{R}$ is a Borel set.

Question

What about *subrings* of $\mathbb{R}$?

Theorem (Edgar-Miller, 2001; Bourgain, 2003)

If $R \subseteq (\mathbb{R}, +, \times, 0, 1)$ is an analytic (i.e. Σ_1^1) *subring* then:

- either $\dim_H(R) = 0$
- or $R = \mathbb{R}$.

This means, R as a subset of $\mathbb{R}$ is an analytic set.

Theorem (R. D. Mauldin, 2016 (R. O. Davies, 1984))

(CH) For every $\alpha \in (0, 1)$ there exists a *subring* $R \subseteq (\mathbb{R}, +, \times, 0, 1)$ such that

$$\dim_H(R) = \alpha.$$

In fact, R is a subfield. It *cannot* be Σ_1^1 .

Mauldin comments: $V=L$ implies the subring can be made Σ_2^1 .

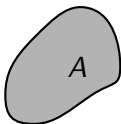
Question

What is the descriptive complexity of such a subring?
Assuming $V=L$, *can it be* Π_1^1 ?

Fact

If $A \subseteq \mathbb{R}$ satisfies $\dim_H(A) = \alpha$ then there exists a G_δ set $D \subseteq \mathbb{R}$

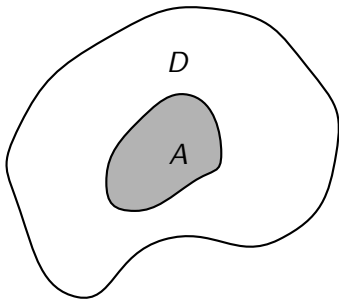
$$\dim_H(D) = \alpha \quad \text{and} \quad A \subseteq D.$$



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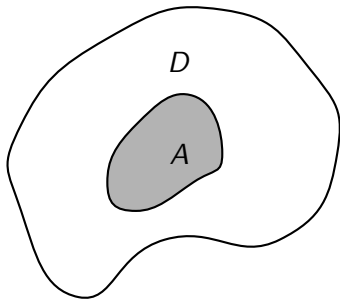
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Fact

If A is not contained in any G_δ set of Hausdorff dimension less than α , then $\dim_H(A) \geq \alpha$.

Generalising an idea of Erdős, K. Kunen, and Mauldin (1981), and A. Miller (1989):

Theorem (Z. Vidnyánszky, 2014)

($V=L$) Let $P \subseteq 2^\omega$ be uncountable Borel. If $F \subseteq \mathbb{R}^\omega \times 2^\omega \times \mathbb{R}$ is $\mathfrak{N}_1^1$ and for every $(A, p) \in \mathbb{R}^\omega \times 2^\omega$, the section $F_{(A,p)}$ is cofinal in $\leq_T$, then there exists a $\mathfrak{N}_1^1$ set $R \subseteq \mathbb{R}$ such that

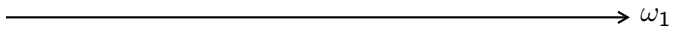
$$P = \{p_\beta \mid \beta < \omega_1\} \quad \text{and} \quad R = \{x_\beta \mid \beta < \omega_1\} \quad \text{and} \quad x_\beta \in F_{(R \upharpoonright_\beta, p_\beta)}.$$

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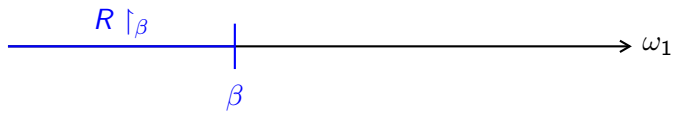
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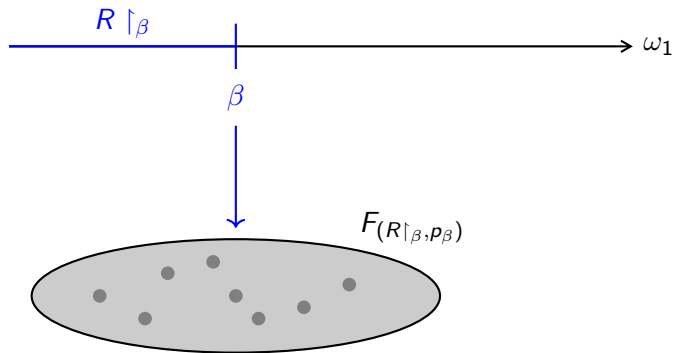
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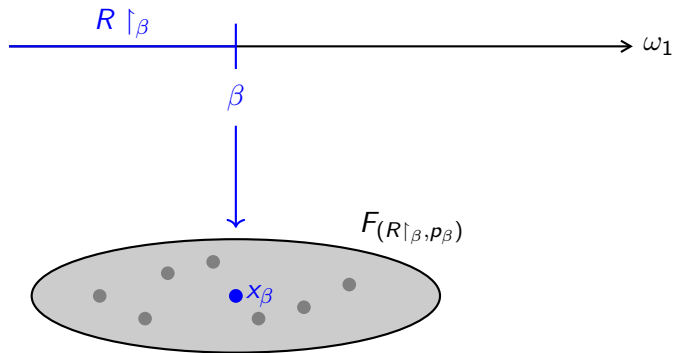
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ω_1







Theorem (R.)

For $\alpha \in (0, 1)$, the set

$$\{c \mid c \text{ is a } G_\delta \text{ Borel code for } A \text{ and } \dim_H(A) \geq \alpha\}$$

is Σ_1^1 -complete.

Corollary

For $\alpha \in (0, 1)$, the set

$$\{c \mid c \text{ is a } G_\delta \text{ Borel code for } A \text{ and } \dim_H(A) < \alpha\}$$

is Π_1^1 -complete.

Proof (hardness)

Reduce a tree T on ω to a suitable Borel code. The idea:

- A path $x \in [T]$ yields an infinite sequence of nested intervals:

$$\langle 1, 2, 3, \dots \rangle \mapsto \langle (0.1, 0.2), (0.12, 0.13), (0.123, 0.124), \dots \rangle$$

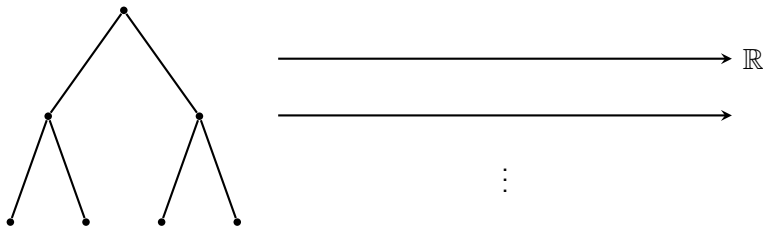
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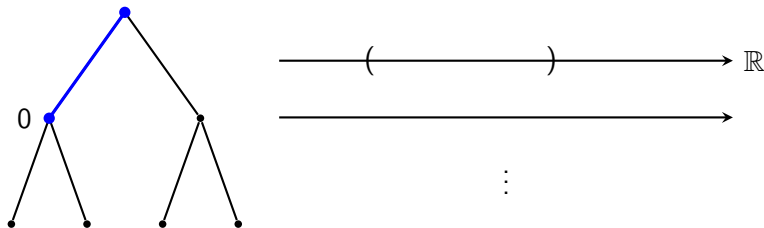
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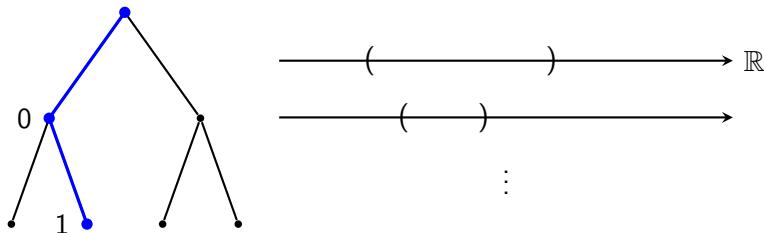
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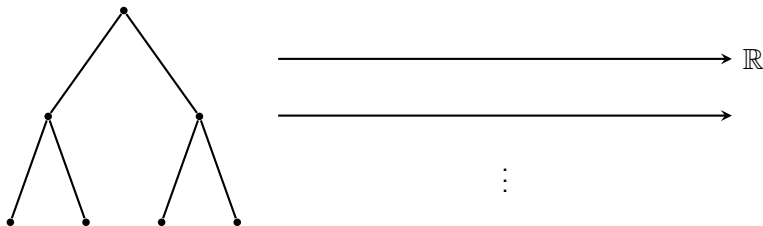
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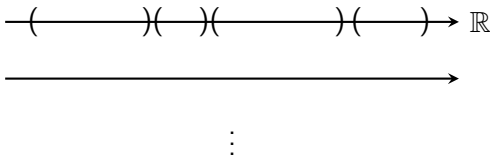
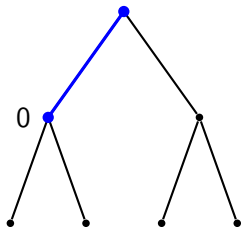
Proposition (R.)

Fix $\alpha \in (0, 1)$. Any infinite path $x \in [T]$ codes a G_δ set $A_x \subset \mathbb{R}$ such that $\dim_H(A_x) \geq \alpha$.



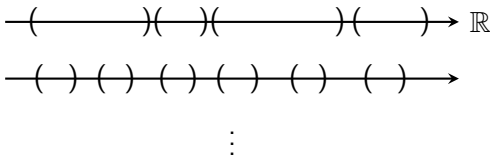
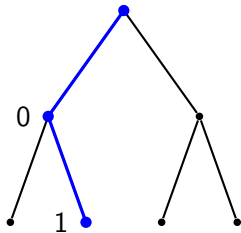
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Proof (membership)

Definition

$B \in 2^\omega$ is a **Hausdorff oracle** for $A \subseteq \mathbb{R}$ if

$$\dim_H(A) = \sup_{x \in A} \dim^B(x).$$

Let A be G_δ . Using the **point-to-set principle**,

$$\dim_H(A) \geq \alpha \iff (\forall n < \omega)(\exists x \in A) \left(\dim^B(x) > \alpha - 2^{-n} \right)$$

where B is a Hausdorff oracle for A .

Fact

If $A \subseteq \mathbb{R}$ satisfies $\dim_H(A) = \alpha$ then there exists a compact set $K \subseteq A$ such that

$$\dim_H(K) = \alpha \quad \text{and} \quad K \subseteq A.$$

Every compact set K is effectively compact [relative to some oracle \$B\$](#) ; then, B is a Hausdorff oracle for K (J. Hitchcock, J. Lutz; D. Stull). Thus,

$$\begin{aligned} \dim_H(A) \geq \alpha &\iff \\ &(\exists K \text{ compact})(\forall n < \omega)(\exists B)(\exists x \in K) \\ &\quad \left(K \subseteq A \wedge B \text{ is Hausdorff for } K \wedge \dim^B(x) > \alpha - 2^{-n} \right). \end{aligned}$$

Proposition (R.)

This clause is Σ_1^1 in the Borel code of A .

What about a “direct” construction?

- What are the conditions?

By the previous theorem, the set of all sufficiently simple G_δ sets is Π_1^1 complete, hence not Borel, hence too complicated.

- Can one build R real by real, by the point-to-set principle?

This is difficult to apply since we must close under $+$ and $\times$. And since $\mathbb{Q} \subset R$, we cannot make use of compactness.

- Even then, we add countably many reals at every step.

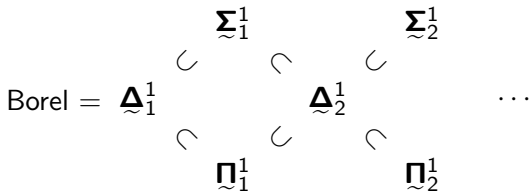
For every x we enumerate into our subring, we must close under all ring-theoretic operations, which yields (under CH) countably many reals to add. A Π_1^1 basis B gives a Σ_2^1 subring R :

$$x \in R \iff (\exists y \in B)(x \text{ is generated by } y).$$

Question

Is there a subring $R \subseteq (\mathbb{R}, +, \times, 0, 1)$ with $\dim_H(R) \neq 0, 1$?

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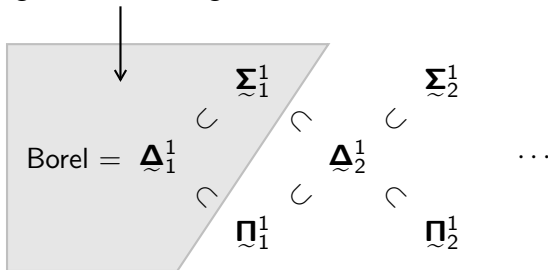


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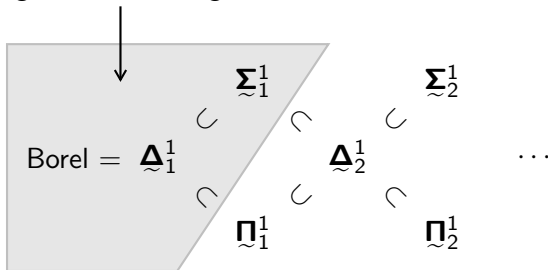


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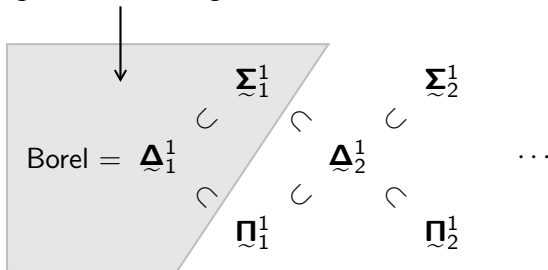


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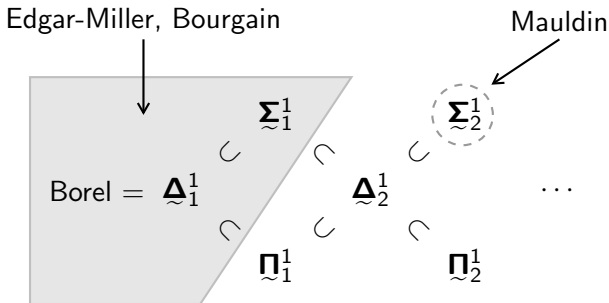
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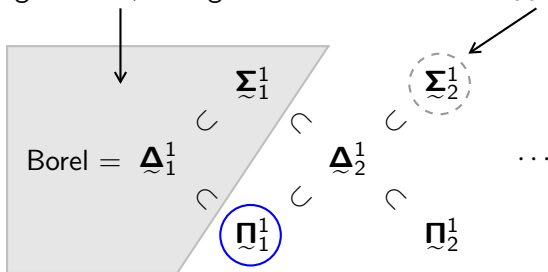
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Mauldin



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*A. Montalbán on Martin's conjecture
NAMS 66(8), 2019*

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